

Question	Answer	Marks
3(a)	Evidence of use of area below the line Distance = area under the graph = $\frac{1}{2}(6+3.8)8 = 39.2 \text{ m}$ Allow $39 \pm 0.5 \text{ m}$ (if greater than 1 but less than 2 m, allow 1 mark of the two)	[1] [1]
3(b)(i)	$\Delta K.E. = \frac{1}{2}m(v_f^2 - v_i^2) = \frac{1}{2}(95)(3^2 - 6^2) = -1280 \text{ J}$	[1]
3(b)(ii)	$\theta = \sin^{-1}(1/30) = 1.91^\circ$ $\Delta G.P.E. = mg\Delta h = (95)(9.81)(39.2 \sin 1.91^\circ) = 1220 \text{ J}$ Values will vary based on answer in (a)	[1] [1]
3(c)	Energy delivered to the bicycle = gain in GPE + gain in KE + energy used to overcome resistive forces Energy used to overcome resistive forces = $\frac{75 \cdot 8 - 1220 + 1280}{2} = 660 \text{ J}$	[1]
3(d)(i)	(Gradient of the graph gives acceleration of the cyclist and bicycle.) <u>The acceleration starts off at zero and increases in magnitude. The direction of the acceleration is opposite to the motion.</u>	[1]
3(d)(ii)	Since <u>air resistance decreases as speed decreases</u> and the component of the weight along the slope constant, <u>hence that the driving force by the cyclist moves the system upslope must be decreasing</u> with time so that the net force/acceleration downslope will have an increasing magnitude. M1 – correct reasoning, A1 – correct deduction about the driving force.	[1] [1]
	Max Marks	10

Question	Answer	Marks
4(a)(i)	$v = r\omega$	